

TITAN-X

Stratospheric Persistent Aerial Platform

Indefinite-endurance free-flight aerostat. 20,000 lb modular payload. California firefighting first; metro disaster and infrastructure nodes next. Planning estimates for qualified parties. Not an offer to sell securities.



California firefighting configuration — retardant + water delivery.

1. Executive summary

TITAN-X is a free-flight stratospheric aerostat in the altitude band between conventional aircraft and satellites. Helium buoyancy plus solar power keep it aloft indefinitely. Gondolas and underbelly modules swap so one hull family serves firefighting, disaster, communications, observation, rockoon loft, and selected dual-use missions.

Investment thesis: pathfinder validation → first full-scale hull → California firefighting concentration + metro pre-positioning → planning **12-unit** fleet by 2030. Pre-positioning converts response from days into hours. Civil utilization underwrites year-round availability; dual-use configurations share the same hull and cost base.

\$13M Unit CAPEX (fleet volume)	\$156M 12-unit fleet CAPEX	\$2.4M Annual OPEX / unit
\$200K Monthly break-even / unit	17.8 : 1 Planning 10-yr fleet ROI	20,000 lb Net modular payload

2. Platform at a glance

Parameter	Value	Notes
Size	480 ft × 200 ft	Swept manta planform
Net payload	20,000 lbs	≈ school-bus weight
Altitude range	500 – 130,000 ft	Tree-top to lower stratosphere
Endurance	Indefinite	Solar + buoyancy; no combustion for loiter
Architecture	Fully modular	Gondola / underbelly module swap
Rockoon loft	~100,000 ft	Propellant savings; platform recovers

2.1 Core advantages

- Indefinite endurance — solar power is sufficient for continuous station-keeping
- School-bus-class payload with rapid mission-module swap on one hull family
- Near-zero operational carbon for persistent missions
- Rockoon loft plus selected dual-use configs share the civil cost base

2.2 How the system works

Lift is helium; electric/solar energy covers station-keeping, sensors, and modules. Firefighting tanks one week, communications the next — without a second fleet. The 500–130,000 ft band sits between aircraft sortie economics and satellite persistence.

3. Problem and market context

Wildfire and disaster response still depends on scarce surge aircraft. Assets arrive late; coverage collapses with weather, tanker availability, or basing. Satellites persist but cannot deliver mass. Aircraft deliver mass but cannot loiter indefinitely. Cities need **hours-scale** airborne nodes — not ad-hoc charter after the event.

Gap: nothing practical today combines indefinite loiter, 20,000 lb modular payload, rapid mission swap, and near-zero persistent ops carbon in the aircraft–satellite band. TITAN-X is persistent aerial **infrastructure**, not a surge novelty.

3.1 Customer / payer segments

Segment	Primary use	Why TITAN-X
State / regional wildfire authorities	CA firefighting concentration	Sensors + retardant/water + incident command on station
Metro / high-value regions	Disaster & communications nodes	Hours-scale response vs days
Critical infrastructure operators	Pipelines, grids, ports, campuses	Continuous observation + logistics
Maritime / shipping	Domain awareness, ship support	Persistent coverage without ship sortie cost
Public safety	Low-cost drone threat awareness	Wide-area airborne node
Science / weather	High-altitude observation	Long dwell, recoverable platform
Rockoon / launch customers	Stratospheric loft	Propellant savings + recovery
Dual-use / defense (selected)	ISR, relay, forward logistics	Shared civil cost base

4. Regional deployment concept

Preferred model: **pre-positioned regional coverage**, not centralized surge. Response time drops from days to hours. Civil utilization pays for year-round availability. One modular hull shifts among firefighting, disaster, and infrastructure without separate fleets.

4.1 Major cities and high-value regions

Each major metro or high-risk region is assigned one TITAN-X (or a small pair), loaded with modular response packages and able to be on station in hours — disaster response, communications restoration, observation, and logistics.

4.2 California firefighting concentration

California is the natural first stack: fire-behavior / thermal / smoke sensors, retardant or water modules, incident-command comms, and decision-support for commanders. Off-season the same hull swaps to metro, infrastructure, or communications packages so the asset is not stranded on a single mission.

4.3 Smaller complementary platforms

Smaller aerostats cover local observation, relay, weather, and lower-cost regional coverage. Full-size TITAN-X is the primary node; smaller platforms take specialized or lower-intensity tasks.

5. Rockoon and dual-use

Rockoon loft to ~100,000 ft targets propellant savings for responsive access, with platform recovery after launch. Dual-use defense options (ISR, relay, forward logistics) are selected configurations on the civil hull — not a separate product line — so civil utilization helps carry the cost base.

6. Development path and capital staging

Capital is staged: **pathfinder** → **first full-scale hull** → **initial operational units** → **fleet scale-up**. Gate heavy fleet CAPEX on endurance demo, modular swap demo, and first paid availability or mission contract.

Phase	Years	Milestone / exit
A. Pathfinder	2026	Endurance, station-keeping, modular swap demonstrated
B. First full-scale hull	2026–2027	Operational unit class
C. Initial nodes	2027–2028	CA firefighting layer + first metro nodes
D. Fleet scale-up	2028–2030	Toward planning 12-unit fleet + complementary platforms



Regional pre-positioning — one modular hull family, persistent high-altitude operations.

7. Unit economics

Item	Planning value
Unit CAPEX (fleet volume)	\$13.0M
12-unit fleet CAPEX	\$156M
Annual OPEX per unit	\$2.4M
Break-even utilization	~\$200K / month (covers OPEX)
Mature utilization target (base)	\$8–12M / unit / year
Contribution at \$10M rev / \$2.4M opex	~\$7.6M / unit / year

7.1 Illustrative mature revenue mix

Stream	Share of unit revenue	Role
Regional standby / availability	35–45%	Underwrites year-round presence
Wildfire / disaster mission fees	20–30%	High intensity, seasonal
Infrastructure / maritime / public safety	15–25%	Steady civil utilization
Rockoon / science / specialty	5–15%	High-margin spikes
Dual-use / defense options	0–20%	Selected configs; shared hull

8. Financial plan 2026–2030

USD millions. The planning 17.8 : 1 ten-year fleet ROI is a long-horizon target on the stood-up fleet — not a 2030 cash guarantee. Early hulls cost more than the \$13M volume target; the model includes learning-curve premium.

8.1 Fleet build (base)

	2026	2027	2028	2029	2030
Full-size units EOY	0*	1	3	6	12
CA firefighting-primary	—	1	2	3	4
Metro / regional nodes	—	0	1	3	8

*Pathfinder year — endurance, station-keeping, modular swap validation.

8.2 Capital expenditure (base, \$M)

	2026	2027	2028	2029	2030	Total
Pathfinder + engineering	18	6	4	3	2	33
Hull production (learning curve)	—	18	30	42	78	168
Ground / ops infrastructure	4	6	8	10	12	40
Total CAPEX	22	30	42	55	92	241

8.3 Operating model (base, \$M)

	2026	2027	2028	2029	2030
Avg units in service	0	0.5	2.0	4.5	9.0
Revenue	0.5	4	18	42	90
Direct OPEX	1.5	3.5	7	13	24
Program / G&A	4	5	7	9	11
EBITDA (planning)	(5.0)	(4.5)	4	20	55

2030 check: ~\$90M revenue on ~9 average units ≈ **\$10M per unit-year**, above the \$2.4M OPEX break-even. Cumulative CAPEX 2026–2030 ≈ \$241M; cumulative revenue ≈ \$154.5M during build years.

8.4 Scenario fan — 2030

Case	Units	Revenue	EBITDA	Note
Conservative	6	\$45M	\$18M	Slower production; CA + few metros
Base	12	\$90M	\$55M	Matches summary fleet target
Upside	12+	\$130M	\$85M	Strong dual-use + rockoon + multi-state wildfire

9. Competition and differentiation

Alternative	Strength	Limitation vs TITAN-X
Tanker / scooping aircraft	Proven mass delivery	No indefinite loiter; weather & basing limits
Helicopters	Precision	Low payload-hours; expensive persistence
LEO / GEO satellites	Persistent eyes	No mass delivery; limited logistics
Tethered aerostats	Persistent local	Site-bound; lower payload / mobility
HAPS aircraft concepts	High altitude	Rarely combine 20k-lb modularity + rockoon + dual-use

Differentiation: indefinite solar/buoyancy endurance × 20,000 lb modularity × regional pre-positioning × shared civil–defense cost base.

10. Use of funds and milestones

Use	Share	Purpose
Pathfinder + first full-scale hull	35–45%	Technical de-risk
CA firefighting + first metro modules	20–30%	Revenue-bearing configurations
Production tooling / learning-curve	15–20%	Drive unit cost toward \$13M volume
Ops bases, helium, insurance, crew	10–15%	Year-round readiness
Corporate / BD / compliance	5–10%	Contracting with public & private buyers

11. Risks and guardrails

Risk	Guardrail
Airworthiness / airspace	Pathfinder before fleet claims
Helium & ground logistics	OPEX includes top-off; dual-sourcing plan
Early unit cost > \$13M	Learning-curve premium modeled; volume target explicit
Public contracting cycles	Civil commercial utilization underwrites standby
Over-claiming ROI	17.8:1 is planning 10-year fleet figure — not a 2026–2030 cash guarantee
Defense narrative crowding civil	Lead with regional civil utility; dual-use as selected configs

12. The ask

We are engaging venture capitalists and business owners to fund or co-develop:

- Pathfinder validation and first full-scale hull
- California firefighting concentration as the first revenue-bearing mission stack
- Metro pre-positioning nodes that turn TITAN-X into practical infrastructure

Return thesis: stand up the planning 12-unit fleet, clear ~\$200K/month per-unit break-even via availability + mission mix, and pursue the long-horizon 17.8 : 1 planning ROI from persistent dual-use utilization — not from a single surge event.